

How LLMs Are Changing Human Relationships

From Career Guidance to Love and Companionship

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About Me

My Research: AI for Career Education

- I study **how AI can help high school students** choose their future careers
- I use a technology called **SGA (Simulation with Generative Agents)**
 - Students answer questions about their interests and values
 - AI reads the answers and suggests career paths for them

How the system works

- Students write about themselves
- AI generates career path ideas
- Students save or skip each idea

What I found

- 245 high school students tried it
- About **1 in 3** AI suggestions were accepted

Today's Topic

LLMs Are Now Entering a New Area — Relationships

Researchers are using AI in areas that affect people's lives...

-  Finding a romantic partner
-  Fixing relationship problems
-  AI itself becoming a "partner"

Today I will talk about how LLMs are changing love and human relationships

Source: Love First, Know Later: Persona-Based Romantic Compatibility Through LLM Text World Engines, Haoyang Shang, Zhengyang Yan, Xuan Liu

1. Social Background

People Are Getting Lonelier and Lonelier

- **Loneliness is a serious social problem** 🚨
 - The U.S. Surgeon General called loneliness "like a disease" (2023)
 - Young people (age 18–34) feel the loneliest
- **Dating apps are very popular (300+ million users worldwide), but...**
 - Many people still can't find a good partner
- **AI is already part of our daily life**
 - AI assistants and generative AI are now normal for us

CNN (2024): About 1 in 3 young men say they have dated an AI

1. Social Background

What Is an LLM?

LLM = Large Language Model

- An AI that learned from billions of sentences
- It can talk, answer, and explain — like a human
- Examples: ChatGPT, Claude, Gemini

Why can LLMs be used for human relationships?

Point	Old AI	LLM
Talking style	Fixed answers	Natural conversation
Understanding you	Not much	Can read your feelings
Emotion	None	Understands context

2. Problem to be Solved

The Problem with Today's Dating Apps ⚠️

- Apps match people by **age, hobbies, and photos** — this has been the same for 20+ years
- But research says: **real compatibility comes from conversation, not profiles**

What Actually Happens

- Someone looks perfect on the app, but when you talk... it feels different
- Many people quit the app because of this

We need a system that checks compatibility through real conversation

2. Problem to be Solved

Misunderstandings and Fights Are Very Common

- The #1 reason couples break up is miscommunication
- Professional counseling costs a lot of money 💵

People Cannot See Their Own Patterns

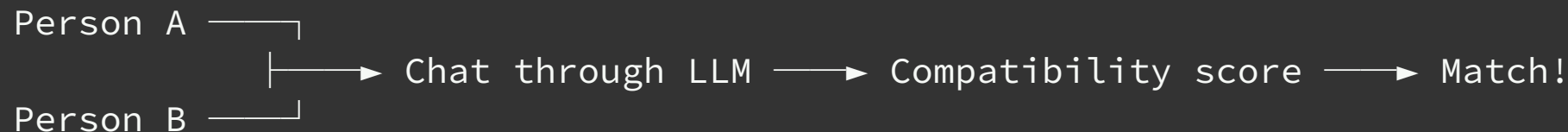
- "Why do we always fight about the same thing?"
- Hard to see from the inside

AI Partners Create New Questions ⚠️

- Is it okay to replace human love with AI?

3. Technical Solution

Idea ①: "Talk First, Then Match"



Steps:

1. Two people have a conversation with LLM as the host
2. LLM watches their talking style, emotions, and response speed
3. Compatibility score is calculated from **real interaction**

Measure compatibility from real conversation, not just a profile

3. Technical Solution

Idea ②: AI Analyzes Couples' Conversations

- A couple gives their LINE or chat history to AI
- LLM finds patterns in their arguments

What LLM finds: 🔍

- "You two always fight about this topic"
- "One person talks too much, the other can't speak"

Then LLM suggests better ways to talk:

- Rewrites the conversation in a healthier way
- Shows examples of good communication ✅

3. Technical Solution

Idea ③: AI as a Companion

What AI companions can do

- Talk with you anytime
- Remembers past conversations
- Never says no to you
- Supports you emotionally

Real examples

- Some people plan anniversaries with AI
- More people say "AI is easier to talk to"

Link to my research

My AI understands a student's **personality and values** to suggest career paths.

AI companions do the same — they understand a person deeply and respond as a "partner."

Same technology, different purpose!

4. Effect of the Solution

Good Effects

For finding a partner:

- Better matches — based on real conversation, not just looks

For couples: 🤝

- Cheap and easy relationship support
- Catch problems early before they get big

For lonely people: 💬

- A safe place to practice social skills

4. Effect of the Solution

Problems and Concerns

Privacy Risk

- Relationship data is the most personal information
- Current laws (like GDPR) were NOT made for this kind of AI

You Might Become Too Dependent on AI

- "I prefer talking to AI than to real people" — this is a real risk

Big Questions We Don't Have Answers for Yet

- If AI gives bad advice and a relationship breaks, who is responsible?

Summary

What I Talked About Today

Section	Key Point
Social Background	Growing loneliness + LLMs becoming part of daily life
Problem	Dating apps match badly; counseling is expensive; no rules for AI love
Technical Solution	Talk-based matching; conflict analysis; AI companions
Effect	Better relationships possible — but privacy and dependency are real risks

LLMs bring new possibilities to human relationships — but we need to think carefully about how we use them

Thank You!

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